

LAB EXPERIMENTS

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Experiment – 15 :

```
import pandas as pd
import numpy as np
# Create a DataFrame
df = pd.DataFrame({
    'A': [1, np.nan, 3, np.nan, 5],
    'B': [np.nan, 2, np.nan, 4, 5],
    'C': [np.nan, np.nan, 3, np.nan, 5],
    'D': [4, 5, np.nan, np.nan, 8]
})
print("Original DataFrame:")
print(df)
# Keep rows having at least 2 NaN values
result = df[df.isna().sum(axis=1) >= 2]
print("\nRows with at least 2 NaN values:")
print(result)
```

```
PS C:\Users\Vedvyass M\Music\Nutri-vision> cd 'c:\Users\Vedvyass M\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle
3 NaN 4.0 NaN NaN
4 5.0 5.0 5.0 8.0

Rows with at least 2 NaN values:
   A  B  C  D
0  1.0 NaN NaN 4.0
1 NaN 2.0 NaN 5.0
2 3.0 NaN 3.0 NaN
3 NaN 4.0 NaN NaN
PS C:\Users\Vedvyass M\Music\Nutri-vision>
```

Experiment – 16 :

```
import pandas as pd

# Create DataFrame

data = {
    'school': ['s001', 's002', 's003', 's001', 's002', 's004'],
    'class': ['V', 'V', 'VI', 'VI', 'V', 'VI'],
    'name': ['Alberto Franco', 'Gino Mcneill', 'Ryan Parkes',
            'Eesha Hinton', 'Gino Mcneill', 'David Parkes'],
    'date_of_Birth': ['15/05/2002', '17/05/2002', '16/02/1999',
                     '25/09/1998', '11/05/2002', '15/09/1997'],
    'age': [12, 12, 13, 13, 14, 12],
    'height': [173, 192, 186, 167, 151, 159],
    'weight': [35, 32, 33, 30, 31, 32],
    'address': ['street1', 'street2', 'street3',
               'street1', 'street2', 'street4']
}

df = pd.DataFrame(data)
print("Original DataFrame:")
print(df)

# Split DataFrame into groups based on school code
grouped = df.groupby('school')
print("\nType of GroupBy object:")
print(type(grouped))
print("\nGroups based on school code:")
for school, group in grouped:
    print("\nSchool:", school)
    print(group)
```

```
Original DataFrame:
  school class      name date_of_Birth  age  height  weight  address
0  s001     V  Alberto Franco  15/05/2002  12    173     35  street1
1  s002     V   Gino Mcneill  17/05/2002  12    192     32  street2
2  s003     VI   Ryan Parkes  16/02/1999  13    186     33  street3
3  s001     VI   Eesha Hinton  25/09/1998  13    167     30  street1
4  s002     V   Gino Mcneill  11/05/2002  14    151     31  street2
5  s004     VI   David Parkes  15/09/1997  12    159     32  street4

Type of GroupBy object:
<class 'pandas.core.groupby.generic.DataFrameGroupBy'>

Groups based on school code:

School: s001
  school class      name date_of_Birth  age  height  weight  address
0  s001     V  Alberto Franco  15/05/2002  12    173     35  street1
3  s001     VI   Eesha Hinton  25/09/1998  13    167     30  street1

School: s002
  school class      name date_of_Birth  age  height  weight  address
1  s002     V   Gino Mcneill  17/05/2002  12    192     32  street2
4  s002     V   Gino Mcneill  11/05/2002  14    151     31  street2

School: s003
  school class      name date_of_Birth  age  height  weight  address
2  s003     VI   Ryan Parkes  16/02/1999  13    186     33  street3

School: s004
  school class      name date_of_Birth  age  height  weight  address
5  s004     VI   David Parkes  15/09/1997  12    159     32  street4
PS C:\Users\Vedvyass M\Music\Nutri-vision>
```

Experiment – 17 :

```
import pandas as pd
# Create DataFrame
data = {
    'school': ['s001', 's002', 's003', 's001', 's002', 's004'],
    'class': ['V', 'V', 'VI', 'VI', 'V', 'VI'],
    'name': ['Alberto Franco', 'Gino Mcneill', 'Ryan Parkes',
            'Eesha Hinton', 'Gino Mcneill', 'David Parkes'],
    'date_of_Birth': ['15/05/2002', '17/05/2002', '16/02/1999',
                     '25/09/1998', '11/05/2002', '15/09/1997'],
```

```

    'age': [12, 12, 13, 13, 14, 12],
    'height': [173, 192, 186, 167, 151, 159],
    'weight': [35, 32, 33, 30, 31, 32],
    'address': ['street1', 'street2', 'street3',
                'street1', 'street2', 'street4']
}

```

```

df = pd.DataFrame(data)
print("Original DataFrame:")
print(df)
# Group by school and calculate mean, minimum and maximum age
result = df.groupby('school')['age'].agg(['mean', 'min', 'max'])
print("\nMean, Minimum and Maximum age for each school:")
print(result)

```

```

PS C:\Users\Vedvyass M\Music\Nutri-vision> c:: cd 'c:\Users\Vedvyass M\Music\Nutri-vision>
ss M\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy'
Original DataFrame:
   school class  name date_of_Birth  age  height  weight  address
0  s001     V  Alberto Franco   15/05/2002   12    173     35  street1
1  s002     V   Gino Mcneill   17/05/2002   12    192     32  street2
2  s003    VI   Ryan Parkes   16/02/1999   13    186     33  street3
3  s001    VI   Eesha Hinton   25/09/1998   13    167     30  street1
4  s002     V   Gino Mcneill   11/05/2002   14    151     31  street2
5  s004    VI   David Parkes   15/09/1997   12    159     32  street4

Mean, Minimum and Maximum age for each school:
      mean  min  max
school
s001    12.5   12   13
s002    13.0   12   14
s003    13.0   13   13
s004    12.0   12   12
PS C:\Users\Vedvyass M\Music\Nutri-vision>

```